FeeFlow ? Technical Design Document

GitHub URL: https://github.com/kavyarathod05/FeeFlow
Live frontend: https://frontend-beta-two-a4xpte0563.vercel.app
Live API: https://feeflow-api-rppm.onrender.com
API health: ‘GET https://feeflow-api-rppm.onrender.com/health‘

ASSUMPTION: Free Render web services sleep after ~15 minutes of inactivity; the first reque

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1. Executive summary

FeeFlow is an MVP SaaS for Company A (Jaipur coaching) that gives parents a clear dues view wit

2. Original business problem

Desktop fee software + Excel due dates + phone chasing. Estimated leakage: late follow-up (~?62

3. Business impact calculations

From the brief (preserved): 2,600 ×?1.2L = ?31.2Cr billed; 2% uncollected ? ?62.4L; staff chas

4. Product scope

In MVP: fee ledger, parent OTP portal, mock pay + receipt, reminder cancel-on-pay, refund c
Roles: Parent, Accounts Executive.
Out of scope: LMS, timetables, admissions CRM.

5. Architecture

React SPA ? Express domain modules ? PostgreSQL (Prisma). In-process cron for mock reminder dis

6. Technology choices

Chosen for typed financial correctness (Decimal), fast MVP UI, Render/Vercel deployability, and

7. Database / ER

26+ relational models including FeePlan, Instalment, Payment, PaymentAllocation, Reminder, Refu

8. Authentication & permissions

- Parent: OTP (hashed, expiry, rate limit); JWT with ‘parentId‘; student access via ‘ParentStud
- Accounts: email/password (bcrypt) + JWT; dashboard/refund/risk/recon restricted.
- Parent ? accounts endpoints ? 403.

9. Fee & instalment logic

‘netFee = base ? discounts‘. Outstanding from successful allocations only. Instalment sum must

10. Payment lifecycle

Link ? mock checkout session ? gateway result ? signed webhook ? transactional allocation ? rec

11. Reminder automation

Configurable T?3 / T0 / T+3/+7/+15 (ASSUMPTION A3). HI/EN templates. Respect consent. Partial:

12. Refund calculation & assumptions

ASSUMPTION A1 (labelled in UI):
‘eligibleNetFee ×remainingDays / totalDays‘, capped at ‘paid ? alreadyRefunded‘. Versioned pol

13. AI risk scoring

Deterministic weighted score from missed instalments, attendance drop, test decline (ASSUMPTION

14. API architecture

REST under ‘/api/*‘ ? auth, students/ledger, instalments/payment-link, payments (checkout/webho

Key auth paths:
- ‘POST /api/auth/parent/request-otp‘
- ‘POST /api/auth/parent/verify-otp‘
- ‘POST /api/auth/accounts/login‘

15. Security & privacy

Zod validation, RBAC, hashed OTP/passwords, webhook HMAC, idempotency keys, audit logs, CORS, r

16. UI / design decisions

Restrained navy/indigo, IBM Plex Sans + Source Serif 4 for amounts. Accounts: sidebar + tables.

Figma: MCP unavailable during build. Design system + 8 screens captured in ‘docs/figma-desi

17. Testing

Vitest: fee, refund, risk, reminder amount rules (18 passing). API smoke verified: pay ? receip

18. Deployment

Layer	Host	URL
Frontend	Vercel	https://frontend-beta-two-a4xpte0563.vercel.app
API	Render (free web)	https://feeflow-api-rppm.onrender.com
Database	Render Postgres (free)	private ‘DATABASE_URL‘

See ‘deployment.md‘. Server binds ‘0.0.0.0:$PORT‘. Empty DB auto-seeds when ‘AUTO_SEED=true‘.

19. Assumptions register

ID	Type	Statement
A1	Business-policy	Illustrative refund formula
A2	Implementation	Eligible net = base ? discounts (? optional non-refundable)
A3	Implementation	Reminder day offsets configurable defaults
A4	Technical	Asia/Kolkata date policy
A5	Implementation	Parent pays specific instalment (not cross-instalment waterfall)
A6	Business-policy	18% GST illustrative on receipts
A7	Technical	Deterministic risk heuristic ? ML
A8	Technical	OTP demo mode
A9	Future/legal	Demo consent ? production DPDP
A10	Data	Brief snapshots preserved alongside live recalculation

20. Intentionally cut

Bulk CSV upload UI, Redis/BullMQ, real SMS/WhatsApp/Razorpay, multi-approver refund workflow, s

21. Future roadmap

Real gateway + messaging, legally reviewed refund policy pack, verifiable parental consent, tra